

BH Project 2026

Chang, Hsin-Chi

Final Project

- A replication study, brain activation
- Research question: Which region of the brain has significant activation during social-cognition tasks?

Learning Goals

- get the experience of doing fMRI data analysis
- understand concept and purpose of each step

Data

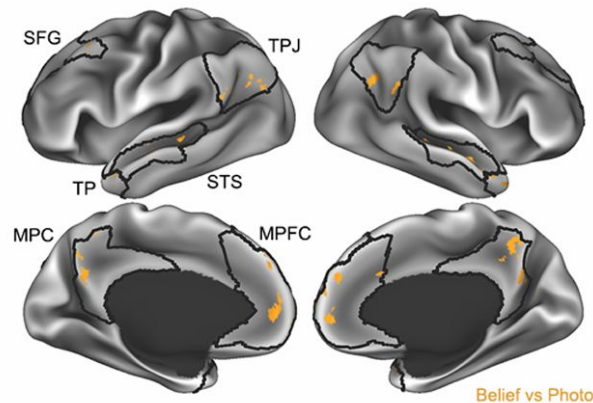
- OpenNeuro ds003814, fMRI
- Social-cognition: Theory of Mind localization
- Raw data, 10 subjects, TD adults

Original paper

- theory of mind tasks

-> define functional ROIs for social reasoning

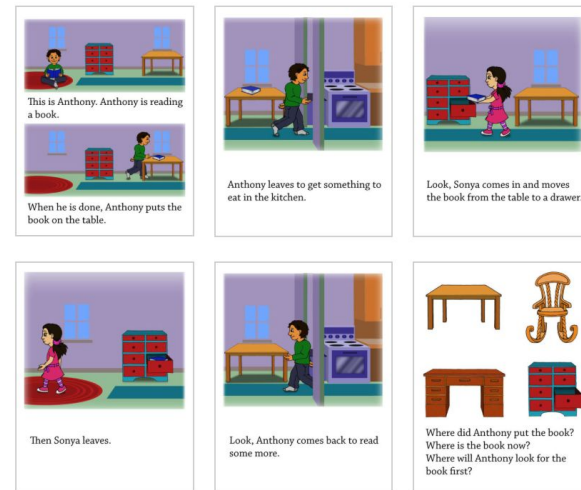
- authors' result: MPC, MPFC, TPJ, STS, SFG, TP



B. Deen, & W.A. Freiwald, Parallel systems for social and spatial cognition reaching the cortical apex, Proc. Natl. Acad. Sci. U.S.A. 122 (44) e2520067122, <https://doi.org/10.1073/pnas.2520067122> (2025).

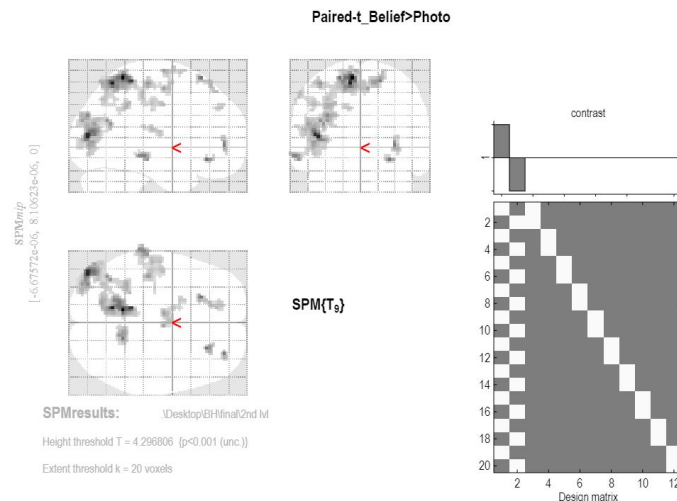
Analysis

- Software: Matlab's SPM
- whole-brain analysis
- **paired t-test, contrast: belief > photo**



Outcome

- p value: < .001
- significant level change: 54
- **7 clusters**
- partially different from the authors'

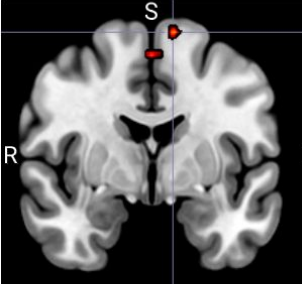
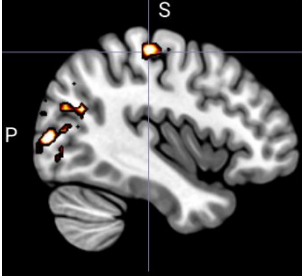
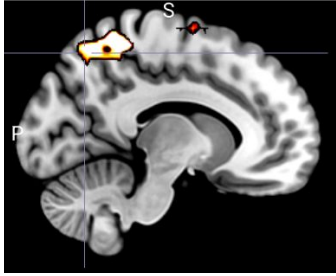
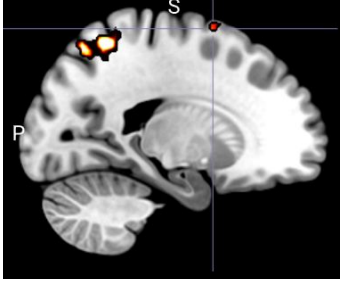
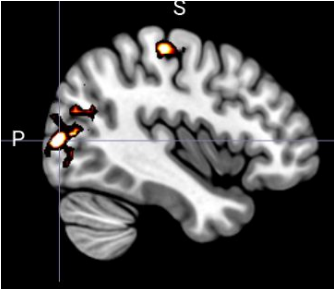
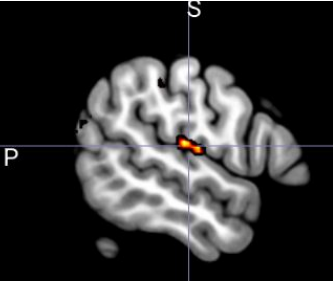
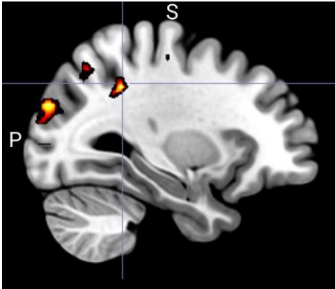


Outline

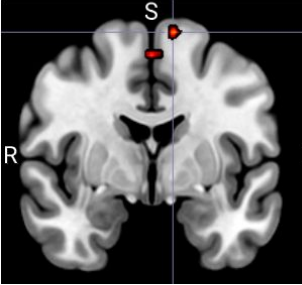
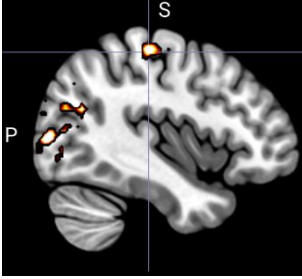
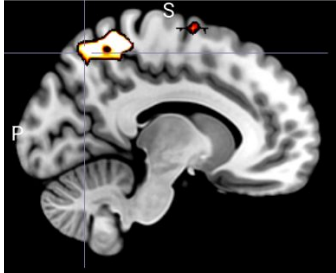
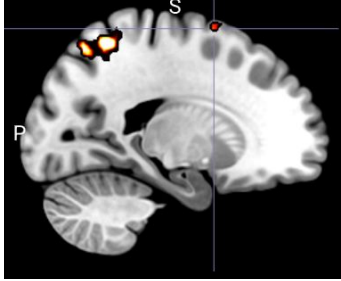
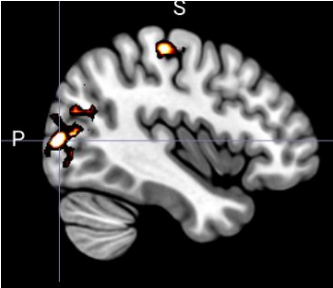
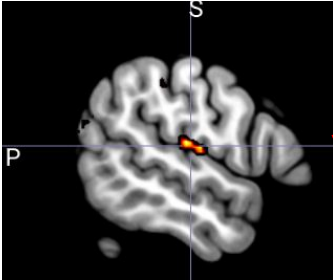
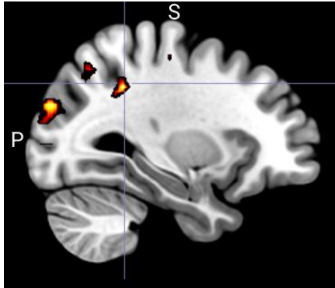
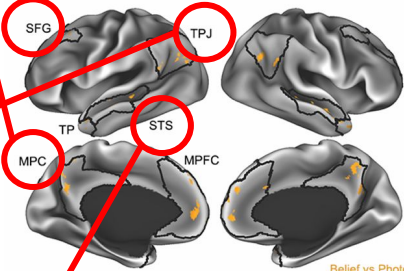
1. The difference between my result and the authors', and what would be the reason
2. How do I preprocess the fMRI data, compared to the original paper
3. What are the problems I ran into during the data analysis process, and how did I fix them
4. What open science operation I have made for reproducibility

The difference between my result and the authors', and what would be the reason

My results

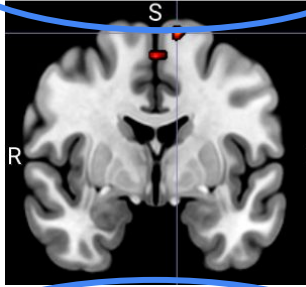
supplementary motor	postcentral gyrus	precuneus	superior frontal gyrus
			
middle occipital gyrus	superior temporal gyrus	inferior parietal lobe	
			

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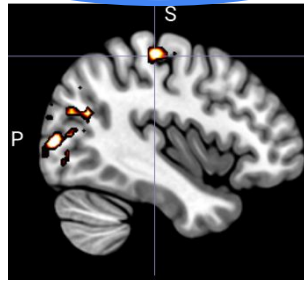
supplementary motor	postcentral gyrus	precuneus	superior frontal gyrus
			
middle occipital gyrus	superior temporal gyrus SFG	inferior parietal lobe TPJ	the authors'
			 Belief vs Photo

My results

supplementary motor



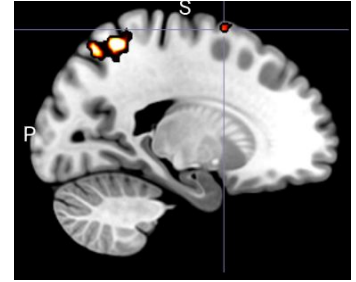
postcentral gyrus



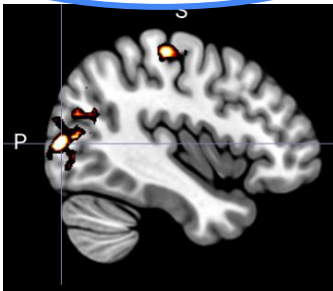
precuneus



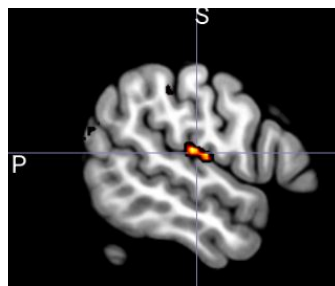
superior frontal gyrus



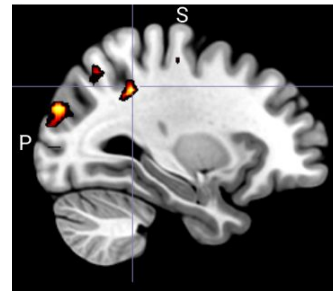
middle occipital gyrus



superior temporal gyrus

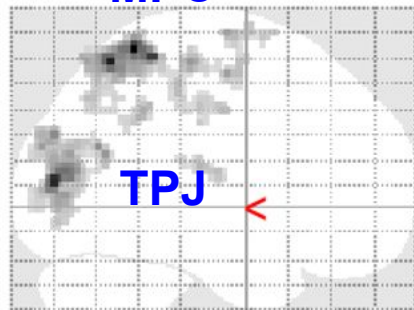


inferior parietal lobe

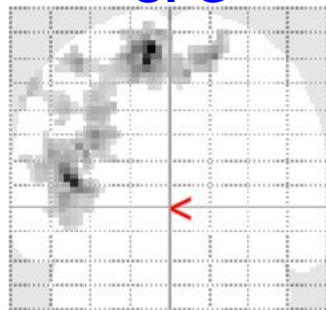


my result

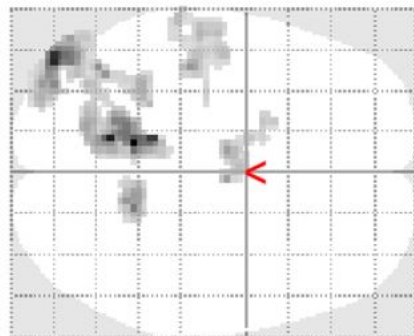
MPC



SFG

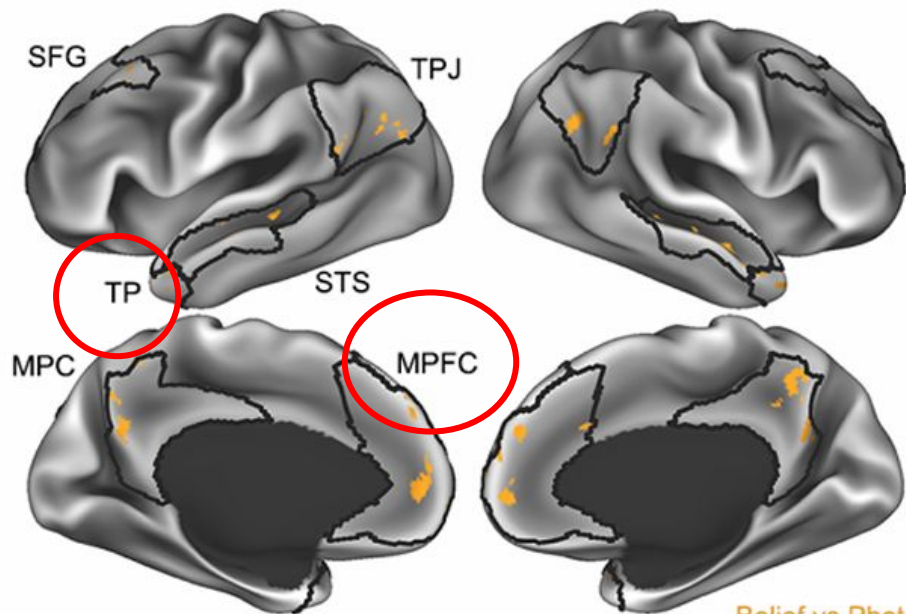


STG



SPM{T₉}

the authors'



Belief vs Photo

Possible cause for difference

- Different way of analysis
1 representative participant
10 subjects' group analysis

- parameter setting
smoothing [5 5 5] -> [7 7 7]

- preprocessing steps
original paper: a lot of denoising

- ROI analysis with AAL
-> check MPFC, TP

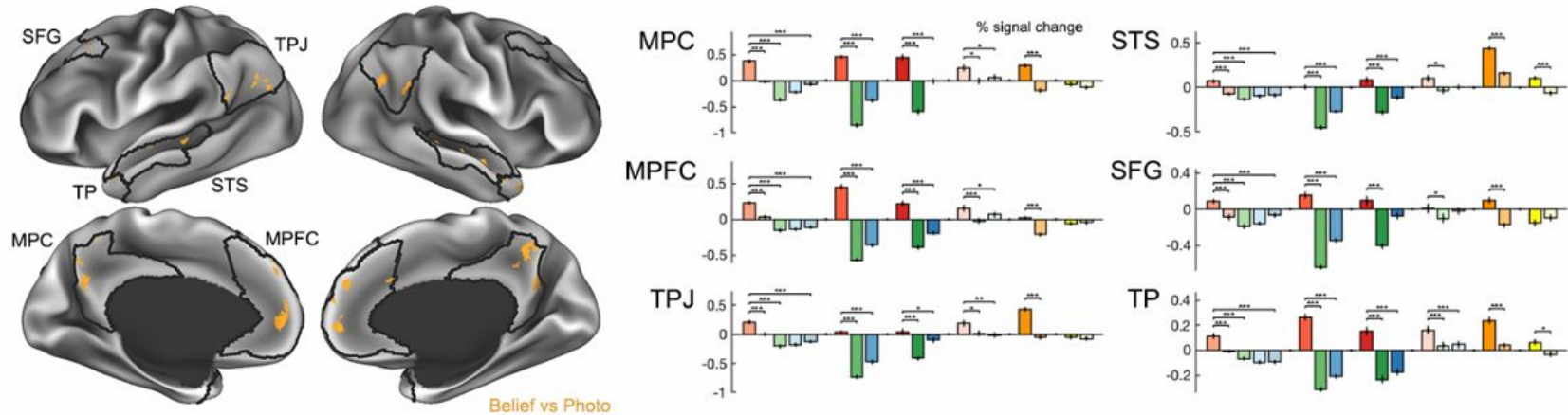
- check my script
-> something wrong?

- need multiecho preprocessing?
-> what else can I do?

Possible cause for difference 1

- Different way of analysis
author: 1 representative participant
mine: 10 subjects' group analysis

- ROI analysis with AAL
-> check TP, MPFC



Possible cause for difference 2

- parameter setting
smoothing [5 5 5] -> [7 7 7]

兩倍法則的數學計算：

標準的高斯平滑（Spatial Smoothing）核心大小（FWHM），最理想的設定是原始體素大小的 2 倍。

- X 軸： $2.4 \times 2 = 4.8$ mm
- Y 軸： $2.4 \times 2 = 4.8$ mm
- Z 軸： $2.5 \times 2 = 5.0$ mm

將這些理論值取最接近的整數，剛好就是完美切中核心的 [5 5 5] ！

Possible cause for difference 2

$$(2.4 * (8/3) \sim 6.4) \rightarrow 7$$

- parameter setting
smoothing [5 5 5] -> [7 7 7]

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- check my batch

-> something wrong?



Possible cause for difference 3

- preprocessing steps
- original paper: a lot of [denoising](#)

Possible cause for difference 3

- the authors'
1. motion estimation
 2. despiking
 3. slice timing correction
 4. single-shot transformation
 5. multiecho-ICA
 6. optimal combination
 7. coregistration
 8. normalization & resampling
 9. spatial smoothing
 10. global signal regression
 11. motion censoring

- preprocessing steps

original paper: a lot of denoising

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1. motion correction
2. slice timing correction
3. coregister T2 to EPI
4. coregister T1 to T2
5. segmentation
6. Dartel create SST
7. Dartel normalize MNI

mine

- preprocessing steps
- original paper: a lot of denoising

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mine

- need multiecho preprocessing?
- > what else can I do?

How do I preprocess the fMRI data,
compared to the original paper

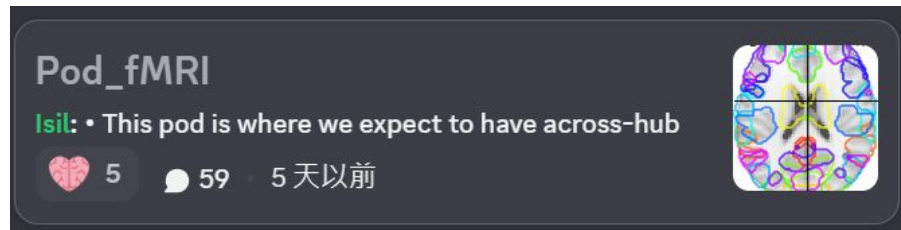
Preprocessing pipeline

1. slice timing
2. motion
correction
3. normalization
4. smoothing

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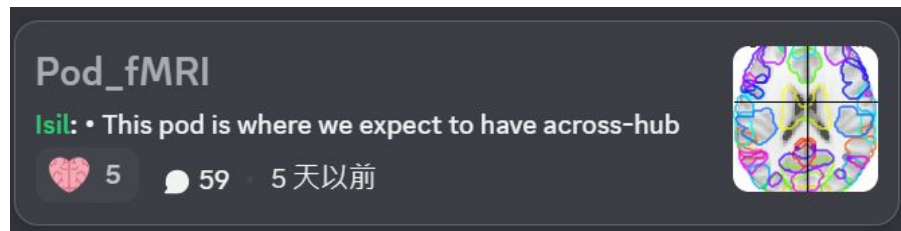
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“scrubbing the time series (removing or interpolating TRs with high motion) before bandpass filtering ”

Preprocessing pipeline

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rather elaborate

“scrubbing the time series (removing or interpolating TRs with high motion) before bandpass filtering ”

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the authors'

removing transient signal spikes

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- avoid image blurring caused by multiple interpolation steps
-

the authors'

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- removing transient signal spikes
- avoid image blurring caused by multiple interpolation steps
- retaining BOLD-related components while removing non-BOLD noise components
-

the authors'

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- removing transient signal spikes
- avoid image blurring caused by multiple interpolation steps
- retaining BOLD-related components while removing non-BOLD noise components
- Multi-echo images were optimally combined using echo-specific weights to maximize BOLD contrast and signal-to-noise ratio.
-

the authors'

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- Removing global brain-wide signal fluctuations
-

the authors'

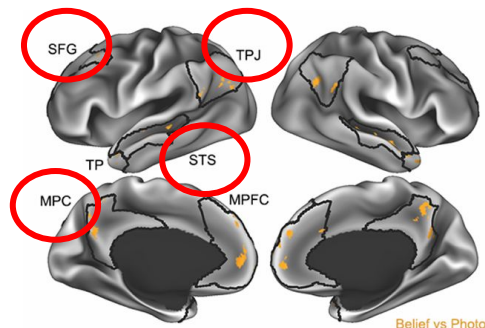
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- removing transient signal spikes
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- retaining BOLD-related components while removing non-BOLD noise components
- Multi-echo images were optimally combined using echo-specific weights to maximize BOLD contrast and signal-to-noise ratio.
- Removing global brain-wide signal fluctuations
- Removing time points with excessive head motion.

the authors' preprocessing

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next step

-> ROI analysis: using AAL atlas to see if
MPFC, TP survive correction

What are the problems I ran into during the data analysis process, and how did I fix them

Problem

- Which run's T1, T2 should I use?
 - Dartel create SST failed
 - Can't open file for writing
 - Dartel normalize MNI
- > untraditional voxel size: 2.4 2.4 2.5

Solution

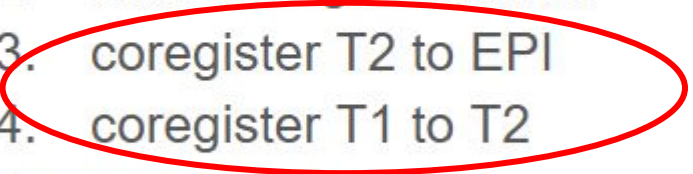
- Quality checking: ventricle, cortex and ghosting artifact
 - copy all the subjects' rc_files to Windows file system
 - call for help in community
- > Gaussian FWHM [7 7 7] ($2.4 \cdot (8/3) \sim 6.4$)

Problem

- Which run's T_1 , T_2 should I use?

Problem

my preprocessing

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Problem

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```
sub-ident(n)/  
├── anat/  
│   ├── run-01/  
│   │   ├── T1w  
│   │   └── T2w  
│   ├── run-02/  
│   │   ├── T1w  
│   │   └── T2w  
│   └── run-03/  
│       ├── T1w  
│       └── T2w
```

Which run's T1, T2
should I use?

Problem

- Which run's T1, T2 should I use?

Solution

- Quality checking: ventricle, cortex, ghosting artifact

Check reg: sub1 **run1** T1w

ventricle

cortex

ghosting

Gray/White Matter
Contrast?

Sharpness
between sulci
and gyri?

motion artifact

Check reg: sub1 **run2** T1w

ventricle

cortex

ghosting

Check reg: sub1 **run3** T1w

ventricle

cortex

ghosting

Check reg: sub1 **ventricle**

run1

run2

run3

Check reg: sub1 **cortex**

run1

run2

run3

probably better
sharpness
between sulci
and gyri?

Check reg: sub1 **ghosting artifact**

run1

run2

probably better

run3

no obvious difference

Check reg: sub2 run1 T1

ventricle

cortex

ghosting

Check reg: sub2 run2 T1

ventricle

cortex

ghosting

Check reg: sub2 run3 T1

ventricle

cortex

ghosting

blurred
ventricle?

preprocessing's segment step could
handle this kind of problem

Strategy

- check reg
- select the run-02 T1

Quality checking:

- sub 1 normal
- sub 2 normal
- sub 3 normal
- sub 4 smaller
- sub 5 bigger
- sub 6 normal
- sub 7 normal
- sub 8 normal
- sub 9 normal
- sub 10 smaller

Problem

- Which run's T1, T2 should I use?

Solution

- Quality checking

-> choose the run-02 T1 for all subjects

Problem

- Which run's T1, T2 should I use?
- Dartel create SST failed
- Can't open file for writing

Solution

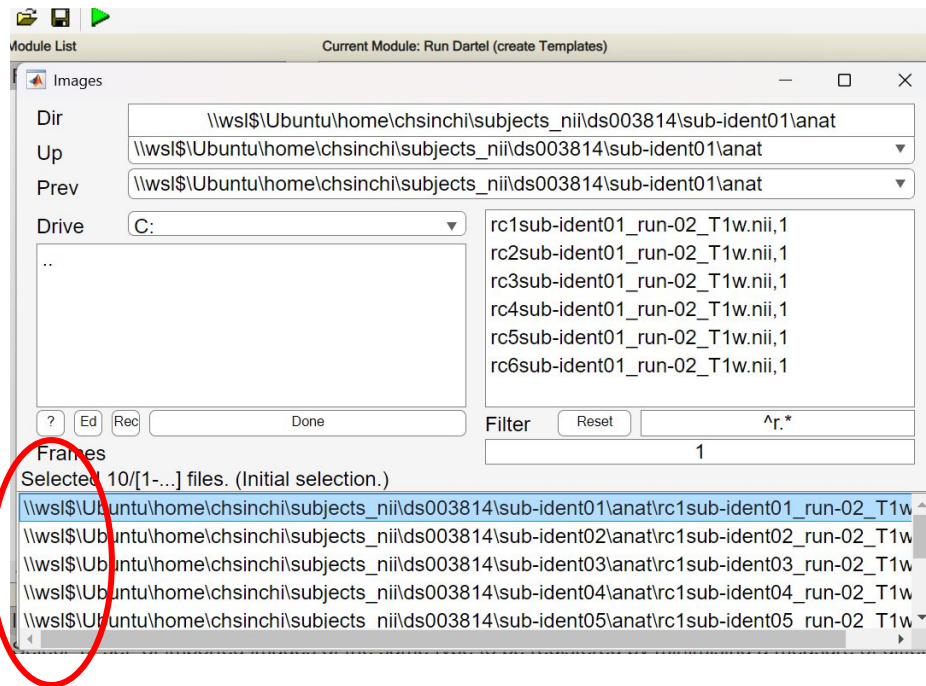
- Quality checking: ventricle, cortex, ghosting artifact

Dartel create SST

- SST: subject specific template
- SST ↔ MNI
 - MNI: Montreal Neurological Institute space
- create the average anatomical template for my subjects

Dartel create SST

- millions of iterative matrix read/write operations
- dataset in WSL file system
- SPM GUI running on native Windows MATLAB
- (\\wsl\$\\) created immense latency
- mat2file write-permission failure
- freezing the MATLAB session



Dartel create SST: failed

Matlab's error message:
Can't open file for writing

```
05-Jun-2026 16:33:12 - Running 'Run Dartel (create Templates)'  
Continuing registration from pre-existing parameters (\\wsl$Ubuntu\home\chsinchi\subjects_nii\ds003814\sub-ident01\anat\u_rc1sub-ident01_run-  
Continuing registration from pre-existing parameters (\\wsl$Ubuntu\home\chsinchi\subjects_nii\ds003814\sub-ident02\anat\u_rc1sub-ident02_run-  
Continuing registration from pre-existing parameters (\\wsl$Ubuntu\home\chsinchi\subjects_nii\ds003814\sub-ident03\anat\u_rc1sub-ident03_run-  
Continuing registration from pre-existing parameters (\\wsl$Ubuntu\home\chsinchi\subjects_nii\ds003814\sub-ident04\anat\u_rc1sub-ident04_run-  
Continuing registration from pre-existing parameters (\\wsl$Ubuntu\home\chsinchi\subjects_nii\ds003814\sub-ident05\anat\u_rc1sub-ident05_run-  
Continuing registration from pre-existing parameters (\\wsl$Ubuntu\home\chsinchi\subjects_nii\ds003814\sub-ident06\anat\u_rc1sub-ident06_run-  
Continuing registration from pre-existing parameters (\\wsl$Ubuntu\home\chsinchi\subjects_nii\ds003814\sub-ident07\anat\u_rc1sub-ident07_run-  
Continuing registration from pre-existing parameters (\\wsl$Ubuntu\home\chsinchi\subjects_nii\ds003814\sub-ident08\anat\u_rc1sub-ident08_run-  
Continuing registration from pre-existing parameters (\\wsl$Ubuntu\home\chsinchi\subjects_nii\ds003814\sub-ident09\anat\u_rc1sub-ident09_run-  
Continuing registration from pre-existing parameters (\\wsl$Ubuntu\home\chsinchi\subjects_nii\ds003814\sub-ident10\anat\u_rc1sub-ident10_run-  
7.82979, 0.259089, 0.0410591, 0.00790892, 0.00303277, 0.00107642, 0.000967744, 0.000658105,  
1 1      710553 0      710553 236.651  
05-Jun-2026 16:34:56 - Failed 'Run Dartel (create Templates)'  
Error using mat2file  
Can't open file for writing:  
    \\wsl$Ubuntu\home\chsinchi\subjects_nii\ds003814\sub-ident01\anat\u_rc1sub-ident01_run-02_T1w_Template.nii  
Check for write permission or whether the directory exists.  
In file "C:\Users\cindy\Downloads\spm@file_array\subsasgn.m", function "subfun" at line 162.  
In file "C:\Users\cindy\Downloads\spm@file_array\subsasgn.m", function "subsasgn" at line 83.  
In file "C:\Users\cindy\Downloads\spm@nifti\subsasgn.m", function "fun" at line 72.  
In file "C:\Users\cindy\Downloads\spm@nifti\subsasgn.m", function "subsasgn" at line 18.  
In file "C:\Users\cindy\Downloads\spm@nifti\subsasgn.m", function "subsasgn" at line 35.  
In file "C:\Users\cindy\Downloads\spm\toolbox\DARTEL\spm_dartel_template.m", function "spm_dartel_template" at line 172.
```

The following modules did not run:
Failed: Run Dartel (create Templates)

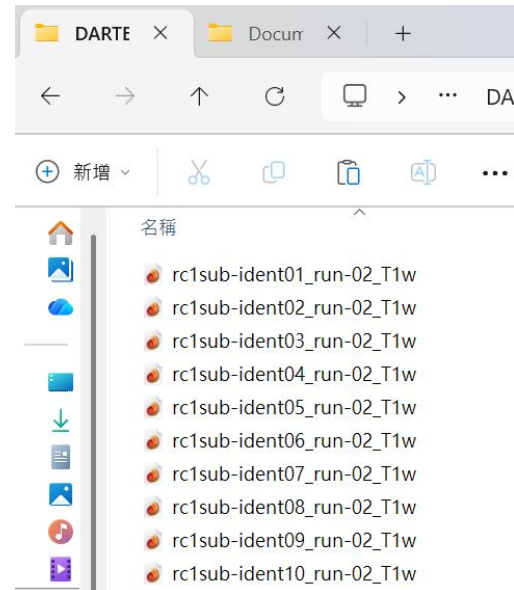
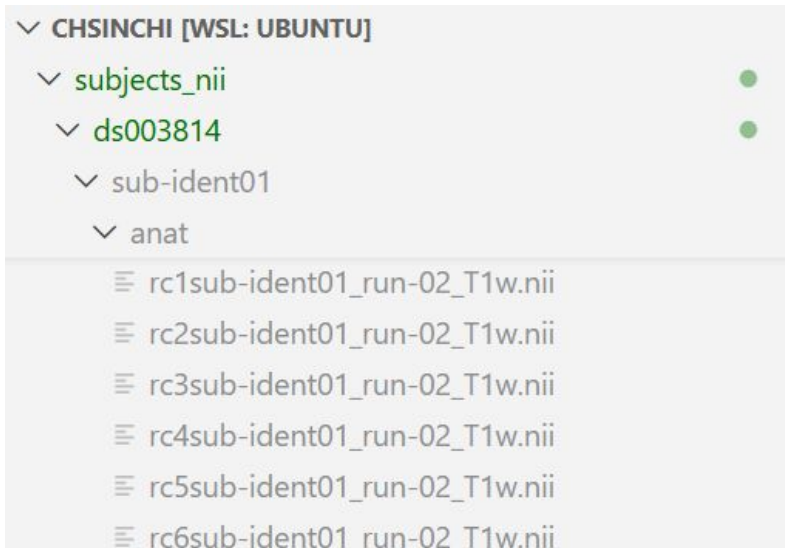
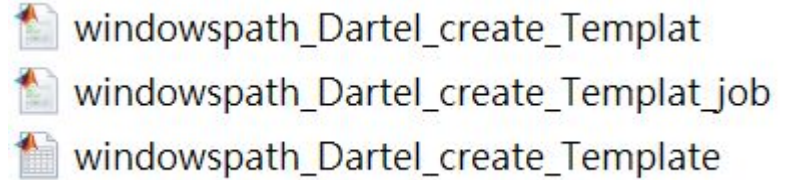
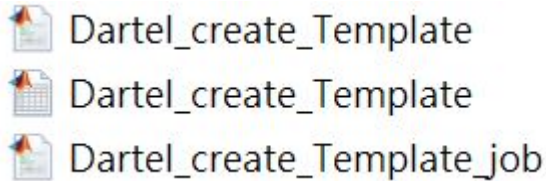
Problem

- Which run's T1, T2 should I use?
- Dartel create SST failed

Solution

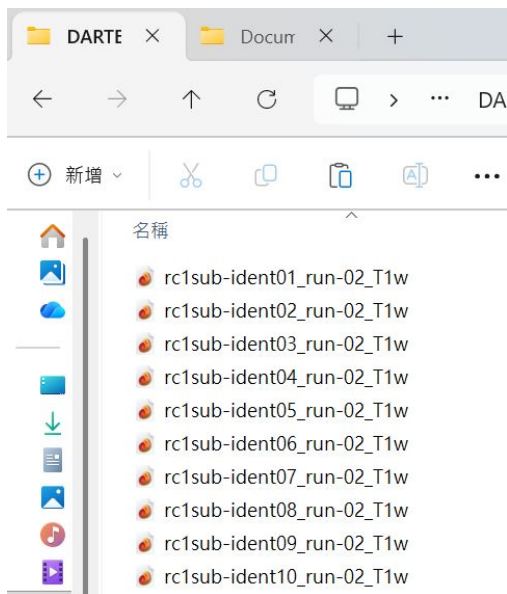
- Quality checking: ventricle, cortex, ghosting artifact
- copy all the subjects' rc_files to Windows file system

copy all the subjects' rc_files to Windows file system



copy all the subjects' rc_files to Windows file system

- ✎ windowspath_Dartel_create_Templat
- ✎ windowspath_Dartel_create_Templat_job
- ✎ windowspath_Dartel_create_Template



my preprocessing

1. motion correction
2. slice timing correction
3. coregister T2 to EPI
4. coregister T1 to T2
5. segmentation
6. Dartel create SST
7. Dartel normalize MNI

file paths are of Windows file system

Problem

- Which run's T1, T2 should I use?
 - Dartel create SST failed
 - Can't open file for writing
 - Dartel normalize MNI
- > untraditional voxel size: 2.4 2.4 2.5

Solution

- Quality checking: ventricle, cortex, ghosting artifact
- copy all the subjects' rc_files to Windows file system

Dartel normalize MNI

- subject ☐ SST ☐ MNI
- including smoothing
- smoothing: for 6 neighbors of a voxel
 - up, down, left, right, forward, backward
 - 26, if corners included
- classical voxel size: [3 3 3]
- default, set to [8 8 8]

Current Module: Normalise to MNI Space

Help on: Normalise to MNI Space

Dartel Template	<-X
Select according to	<-X
Voxel sizes	[NaN NaN NaN]
Bounding box	2x3 double
Preserve	Preserve concentrations
Gaussian FWHM	[8 8 8]

ds003814 dataset

- voxel size: [2.4 2.4 2.5]
- set to [5 5 5]

?

🏆 最推選項： [5 5 5] (黃金中庸之道，完美切中理論值！)

○ 分析：

- 對照我們的兩倍理論值 [4.8, 4.8, 5.0]，四捨五入到整數，剛好就是最完美的 [5 5 5]！
- **優點：**它完美兼顧了「消除 EPI 殘餘鬼影/噪訊」與「保留 2.4mm 高解析度皮質訊號」的雙重需求。對於心智理論 (ToM) 這種高度依賴特定皮質 (如庫 precuneus, TPJ) 的作業 (Task fMRI)，[5 5 5] 能夠提供最剛好的統計檢定力 (Statistical Power)。

+ 問問 Gemini

Flash ▾



Gemini 是 AI，有時可能會出錯。

Problem

- Which run's T1, T2 should I use?
- Dartel create SST failed

-> SPM in Windows, lack of perssion to write file in Linux

- Dartel normalize MNI

-> untraditional voxel size: 2.4 2.4 2.5

Solution

- Quality checking: ventricle, cortex and ghosting artifact
- copy all the subjects' rc1~rc10 nifti files to Windows desktop
- call for help in community ...

Dartel normalize MNI

- classical voxel size: [3 3 3]
 - default, set to [8 8 8]
- ds003814 voxel size: [2.4 2.4 2.5]
 - should set to [7 7 7]



$$(2.4 * (8/3) \sim 6.4) \rightarrow 7$$

- parameter setting
smoothing [5 5 5] -> [7 7 7]

~~兩倍法則的數學計算：~~

~~標準的高斯平滑 (Spatial Smoothing) 核心大小 (FWHM)，最理想的設定是原始體素大小的 2 倍。~~

~~◦ X 軸：2.4 × 2 = 4.8 mm~~

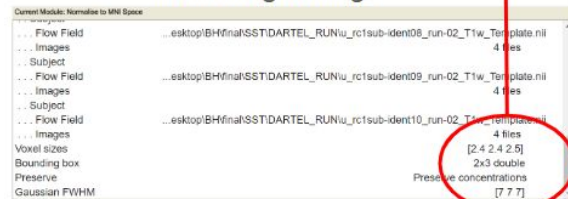
~~◦ Y 軸：2.4 × 2 = 4.8 mm~~

~~◦ Z 軸：2.5 × 2 = 5.0 mm~~

~~將這些理論值取最接近的整數，剛好就是完美切中核心的 [5 5 5] !~~

- check my batch

-> something wrong?



Problem

- Which run's T1, T2 should I use?
 - Dartel create SST failed
 - Can't open file for writing
 - Dartel normalize MNI
- > untraditional voxel size: 2.4 2.4 2.5

Solution

- Quality checking: ventricle, cortex and ghosting artifact
 - copy all the subjects' rc_files to Windows file system
 - call for help in community
- > Gaussian FWHM [7 7 7] ($2.4 * (8/3) \sim 6.4$)

Problem & Solution

- Which run's T1, T2 should I use?

☐ check reg

- Dartel create SST failed

☐ move files to Windows system

- Dartel normalize MNI

☐ call for in BH community

my conclusion

need quality checking

file and software better in the same system

formal training and community matters ^^

Problem & Solution

- Which run's T1, T2 should I use?

☐ check reg

- Dartel create SST failed

☐ move files to Windows system

- Dartel normalize MNI

☐ call for in BH community

my conclusion

need quality checking

file and software better in the same system

formal training and community matters ^^

☐ open science

What open science operation I have made for
reproducibility

my repo

 README  MIT license

Brain Activation during ToM task

This is a replication study of the ToM (theory of mind) localizer using fMRI.

Research question

Which region of the brain has significant activation during the ToM task?

Data Analysis

All the fMRI data preprocessing and analysis are conducted in Matlab's SPM with `.m` script provided.

Get data

See: [get_data.txt](#)

Environment

- Windows11
- Matlab R2025b
- SPM25

Packages

No packages published
[Publish your first package](#)

Contributors 1



r14227123-cloud Chang, Hsin-Chi

Languages



https://github.com/r14227123-cloud/Hsin-Chi_Project

File structure

Hsin-Chi_Project / 2. Documents / file_structure.txt 

 r14227123-cloud add file structure

Code Blame

```
1  | 1. Command_Files/
2  |   └─get_data.txt
3  |
4  | 2. Documents/
5  |   └─file_structure.txt
6  |   └─event-related_design_example
7  |   └─useful_mri_info.txt
8  |
9  | 3. Original_Data/
10 |   └─data_structure.txt
11 |   └─Step1~5
12 |       └─ten_subjects_preprocessing.m
13 |       └─ten_subjects_preprocessing_job.m
14 |       └─for_loop_version
15 |           └─tom_preprocessing_10sub.m
16 |           └─tom_preprocessing_10sub_job.m
17 |   └─Step6
18 |       └─windowspath_Dartel_create.m
19 |       └─windowspath_Dartel_create_job.m
20 |   └─Step7
```

```
20 |   └─Step7
21 |       └─Dartel_normalize_MNI.m
22 |       └─Dartel_normalize_MNI_job.m
23 | 4. Analysis_Data/
24 |   └─Step8
25 |       └─ten_subjects_1stlvl_spec.m
26 |       └─ten_subjects_1stlvl_spec_job.m
27 |       └─multiple_condition.m
28 |   └─Step9
29 |       └─ten_subjects_1stlvl_est.m
30 |       └─ten_subjects_1stlvl_est_job.m
31 |   └─Step10
32 |       └─ten_subjects_1stlvl_contrast.m
33 |       └─ten_subjects_1stlvl_contrast_job.m
34 |   └─Step11
35 |       └─second_lvl_spec.m
36 |       └─second_lvl_spec_job.m
37 |   └─Step12
38 |       └─second_lvl_est.m
39 |       └─second_lvl_est_job.m
```

how do I ...

- get the ds0003814 data
- preprocessing
- subject-level analysis
- group-level analysis
- results

 r14227123-cloud update event-related example		
	Analysis_Data	add .m scripts
	Command_Files	add Ubuntu terminal command
	Documents	update event-related example
	Original_Data	update preprocessing scripts
	.gitignore	Create .gitignore
	LICENSE	Initial commit
	README.md	update README
	environment.yml	Add environment file

how do I ...

- get the ds0003814 data

- preprocessing

- subject-level analysis

- group-level analysis

- results

 r14227123-cloud update event-related example		
	Analysis_Data	add .m scripts
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	Documents	update event-related example
	Original_Data	update preprocessing scripts
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	LICENSE	Initial commit
	README.md	update README
	environment.yml	Add environment file

how do I ...

- get the ds0003814 data
 - Command_Files/
 - terminal command

[Hsin-Chi_Project](#) / [Command_Files](#) / [get_data.txt](#) 



r14227123-cloud add Ubuntu terminal command

Code

Blame

```
1   conda activate datalad_env
2
3   cd Hsin-Chi_Project
4
5   datalad clone https://github.com/OpenNeuroDatasets/ds003814.git
6
7   cd ds003814
8
9   datalad get sub-ident*/func/sub-ident*_task-tomloc2_run-0{1..4}_e
10
11  git annex unlock
12
13  find . -name "*.nii.gz" -exec gunzip {} \;
```

how do I ...

- preprocess the original data
 - Original_Data/
 - take a glance at data structure

Hsin-Chi_Project / Original_Data / data_structure.txt

Code

Blame

```
3
4 sub-ident(n)/
5 |---anat/
6 |   |---run-01/
7 |       |--- T1w
8 |       |--- T2w
9 |   |---run-02/      <-- ready for use
10 |       |--- T1w
11 |       |--- T2w
12 |   |---run-03/
13 |       |--- T1w
14 |       |--- T2w
15 |---fmap/      <-- won't use them
16 |   |---AP/
17 |   |---PA/
18 |
19 |---func/
20 |   |---dyloc/
21 |   |   |--- .....
22 |   |   .   .....
23 |   |   .   .....
24 |   |---tomloc2/      <-- ready for use
25 |       |---run-01/
26 |           |--- echo-1
27 |           |--- echo-2
```

r14227123-cloud update preprocessing scripts

Name

- ..
- Step1~5
- Step6
- Step7
- .gitkeep
- data_structure.txt

how do I ...

- preprocess the original data
 - Original_Data/
 - the statement of step1~7 in README

Analysis pipeline

- Preprocessing
 - Step1. motion correction
 - Step2. slice timing correction
 - Step3. coregister T2 to EPI
 - Step4. coregister T1 to T2
 - Step5. segmentation
 - Step6. Dartel create SST
 - Step7. Dartel normalize MNI

Hsin-Chi_Project / Original_Data / 



r14227123-cloud update preprocessing scripts

Name



..



Step1~5



Step6



Step7



.gitkeep



data_structure.txt

Dartel normalize MNI: parameters

- Voxel size: 2.4 2.4 2.5
- Gaussian FWHM [7 7 7] ($2.4 \cdot (8/3) \sim 6.4$)

Dartel normalize MNI: notes

- Template_6 is in Windows
- all the u_rc files for subject 01~10 are in Windows
- all the ar_ files for subject 01~10 are in Linux

Derivatives of preprocessing, for each subject

n run: 1-4

n segment: 1-6

- func/
 - rsub-ident01_task-tomloc2_run-01_echo-2_bold.nii: with realign
 - arsub-ident01_task-tomloc2_run-01_echo-2_bold.nii: with slice timing after realign
 - meansub-ident01_task-tomloc2_run-01_echo-2_bold.nii
 - rp_sub-ident01_task-tomloc2_run-01_echo-2_bold.txt
- anat/
 - c1sub-ident01_run-02_T1w.nii
 - rc1sub-ident01_run-02_T1w.nii
 - mwc1sub-ident01_run-02_T1w.nii
 - iy_sub-ident01_run-02_T1w.nii

how I do ...

- subject-level analysis
- group-level analysis
 - Analysis_Data/

r14227123-cloud update event-related example		
 Analysis_Data		add .m scripts
 Command_Files		add Ubuntu terminal command
 Documents		update event-related example
 Original_Data		update preprocessing scripts
 .gitignore		Create .gitignore
 LICENSE		Initial commit
 README.md		update README
 environment.yml		Add environment file

how I do ...

- subject-level analysis
- group-level analysis
 - **Analysis_Data/**

Hsin-Chi_Project / Analysis_Data /

r14227123-cloud add .m scripts

Name

..

Steo10. 1st_lv_contrast

Step11. 2nd_lv_spec

Step12. 2nd_lv_est

Step8. 1st_lv_spec

Step9. 1st_lv_est

.gitkeep

r14227123-cloud update event-related example

Analysis_Data add .m scripts

Command_Files add Ubuntu terminal command

Documents update event-related example

Original_Data update preprocessing scripts

.gitignore Create .gitignore

LICENSE Initial commit

README.md update README

environment.yml Add environment file

- Subject-level analysis

Step8. first level specification

Step9. first level estimation

Step10. first level contrast

- Group-level analysis

Step11. second level specification

Step12. second level estimation

1st level specification: parameters

- units for design: seconds
- interscan interval: 2 (TR = 2)
- microtime resolution: 48 (48 slices)
- microtime onset: 1 (0 secs: 1, 17, 33 slice)
- **experiment: event-related design**

[Hsin-Chi_Project](#) / [Analysis_Data](#) / [Step8. 1st_lvl_spec](#) / [multiple_condition.m](#) 



r14227123-cloud add multiple_condition.m

Code

Blame

```
1  %% =====  
2  %   .tsv to soa.mat Converter  
3  %   Created for ds003814 Dataset (10 Subjects x 4 Runs)  
4  %% =====  
5  
6  clear; clc;  
7  
8  % 定義資料主目錄  
9  base_dir = '\\ws1$\Ubuntu\home\chsinchi\subjects_nii\ds003814\';  
10  
11 % 10 個受試者與 4 個 Run 的宣告  
12 subject_list = {'01', '02', '03', '04', '05', '06', '07', '08', '09', '10'};  
13 num_runs = 4;  
14
```

1st level s



r14227123-cloud update event-related example

6f27c03 ·

13 lines (13 loc) · 378 Bytes

- units for c
- interscan
- microtime
- microtime
- **experime**
- **multiple c**
 - Analys

Code

Blame

Raw



	1	onset	duration	trial_type
	2	12	14	Photo
	3	38	14	Belief
	4	64	14	Belief
	5	90	14	Photo
	6	116	14	Belief
	7	142	14	Photo
	8	168	14	Photo
	9	194	14	Belief
	10	220	14	Photo
	11	246	14	Belief
	12	272	14	Belief
	13	298	14	Photo

[_spec](#) / [multiple_condition.m](#)

jects x 4 Runs)

\subjects_nii\ds003814\';

'05', '06', '07', '08', '09', '10'];

1st level specification: parameters

- units for design: seconds
- interscan interval: 2 (TR = 2)
- microtime resolution: 48 (48 slices)
- microtime onset: 1 (0 secs: 1, 17, 33 slice)
- experiment: event-related design
- **multiple condition: refer to the .m script**
 - Analysis_Data/ Step8. 1st_lvl_spec/

[Hsin-Chi_Project](#) / [Analysis_Data](#) / [Step8. 1st_lvl_spec](#) / [multiple_condition.m](#) 

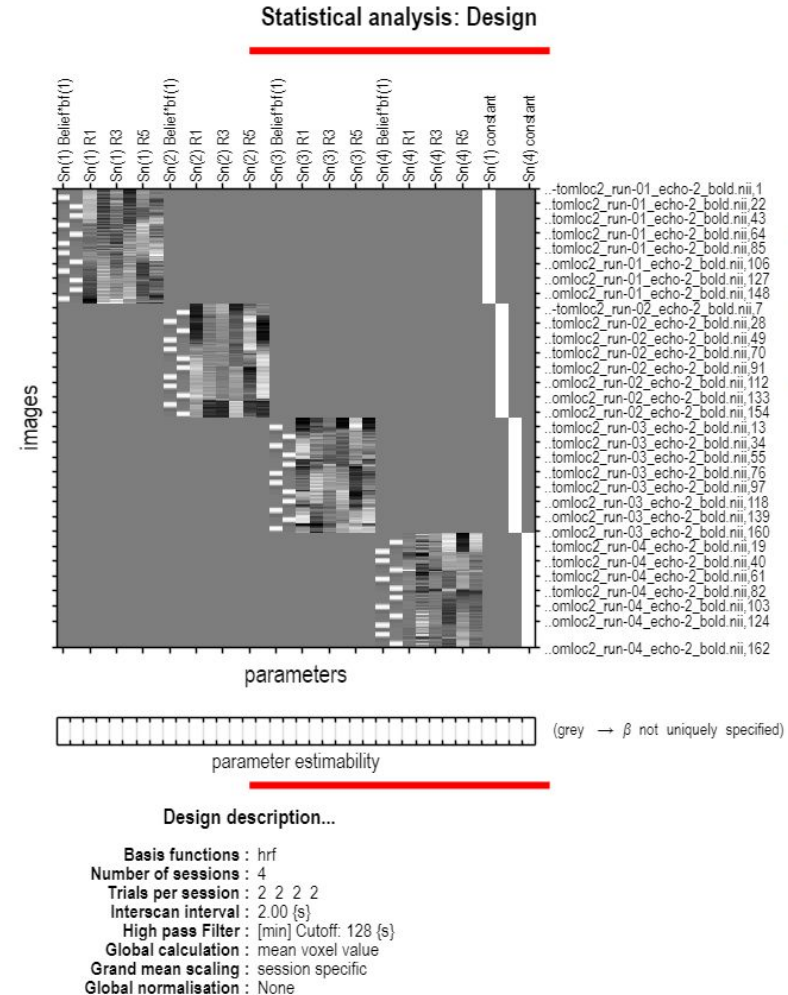
 r14227123-cloud add multiple_condition.m

Code Blame

```
1  %% =====  
2  %   .tsv to soa.mat Converter  
3  %   Created for ds003814 Dataset (10 Subjects x 4 Runs)  
4  %% =====  
5  
6  clear; clc;  
7  
8  % 定義資料主目錄  
9  base_dir = '\\ws1$\Ubuntu\home\chsinchi\subjects_nii\ds003814\';  
10  
11 % 10 個受試者與 4 個 Run 的宣告  
12 subject_list = {'01', '02', '03', '04', '05', '06', '07', '08', '09', '10'};  
13 num_runs = 4;  
14
```

derivative of 1st level specification

- an **SPM.mat** file for each subject

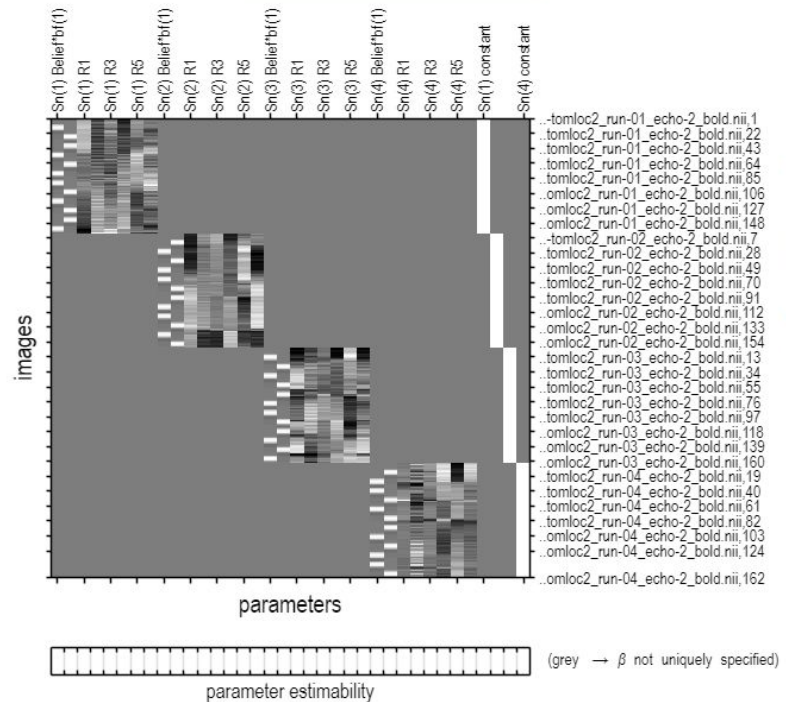


derivative of subject 01

4 runs, for each run:

- Belief
- Photo
- R1
- R2
- R3
- R4
- R5
- R6
- Constant

Statistical analysis: Design



Design description...

Basis functions : hrf
Number of sessions : 4
Trials per session : 2 2 2 2
Inter-scan interval : 2.00 (s)
High pass Filter : [min] Cutoff: 128 (s)
Global calculation : mean voxel value
Grand mean scaling : session specific
Global normalisation : None

1st level estimation: parameter

- write residuals: no (I'm not doing functional connectivity analysis)
- method: classical (maximum likelihood estimation)

derivatives of 1st level estimation

- for each subject:
 - **beta**_0001.nii
 - **mask**.nii: 遮罩
 - **ResI**_0001.nii
 - **ResMS**.nii: residual
 - **RPV**.nii: multiple comparison

n coefficients: 1~36

n ? : 1~64

1st level contrast: parameter

- 2 contrasts: Belief, Photo
- Weight vector:
 - Belief: 1 0 0 0 0 0 1 0 0 0 0 0 1 0 0 0 0 0 1 0 0 0 0 0
 - Photo: 0 1 0 0 0 0 0 1 0 0 0 0 0 1 0 0 0 0 0 1 0 0 0 0

derivatives of 1st level contrast

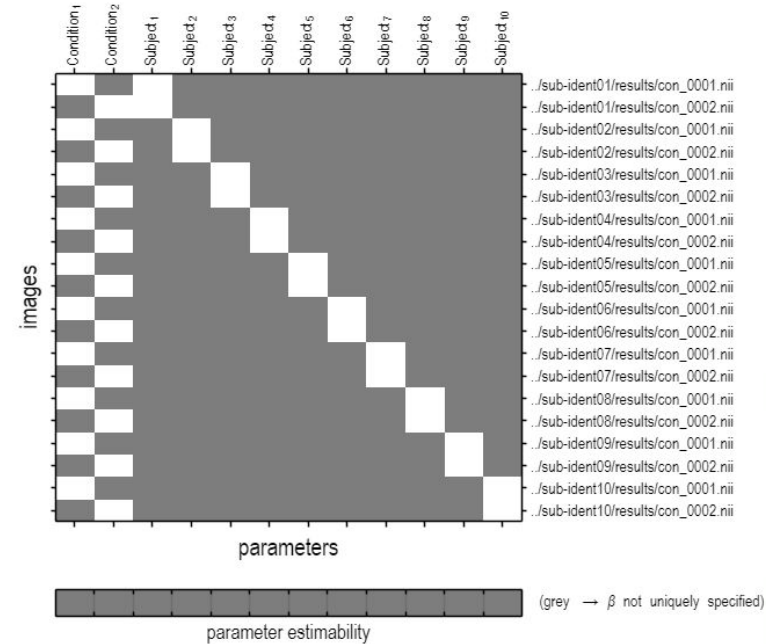
- for each subject:
 - `con_0001.nii`: Belief
 - `con_0002.nii`: Photo
 - `spmT_0001.nii`: Beliefi
 - `spmT_0002.nii`: Photo

2nd level specification: parameters

- **paired-t test** n subjects: 1~10
- 10 pairs
- for each pair
 - scans[1,2]: sub-ident01/results/con_0001.nii, sub-ident01/results/con_0002.nii
- grand mean scaling: no
- ANCOVA: no
- left other settings as default

derivative of 2nd level specification

- **SPM.file**
- images: 20 contrasts nifti file
- parameters: 2 conditions, 10 subjects



Design description...

Design : Paired t-test
Global calculation : omit
Grand mean scaling : <no grand Mean scaling>
Global normalisation : <no global normalisation>
Parameters : 2 condition, +0 covariate, +10 block, +0 nuisance
 12 total, having 11 degrees of freedom
 leaving 9 degrees of freedom from 20 images

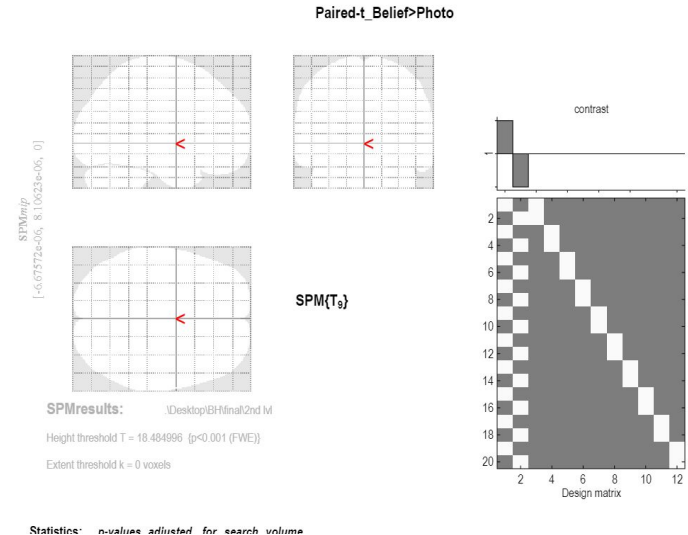
2nd level estimate

- SPM.file of 2nd level specification

derivative of 2nd level estimate

- beta_0001.nii
- mask
- ResMS
- RPV

n coefficients: 1~12



Results

- **check results**
 - Documents/

[Hsin-Chi_Project](#) / [Documents](#) / results figure /

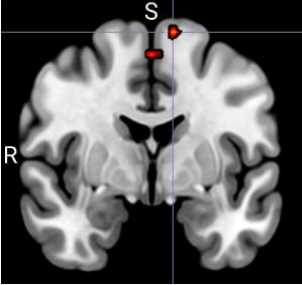
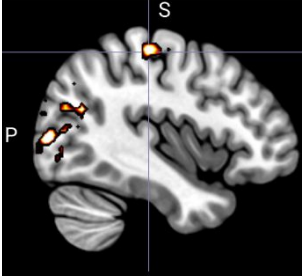
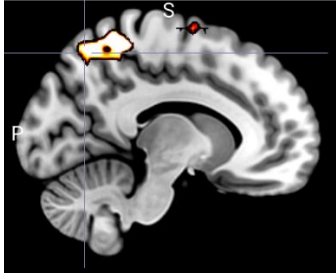
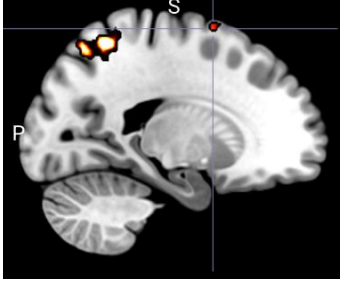
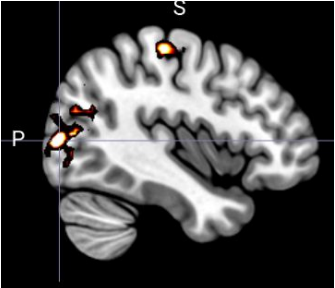
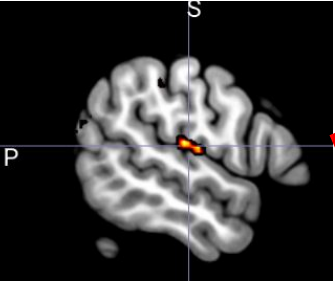
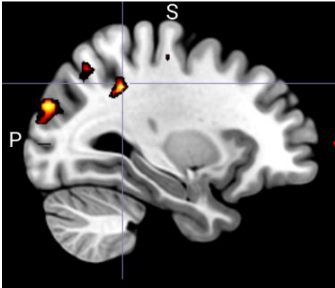
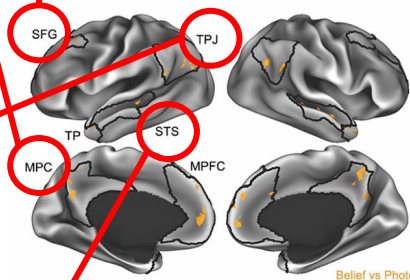


r14227123-cloud update README

Name
 ..
 significantlevelchange_54_001.png
 significantlevelchange_54_002.png

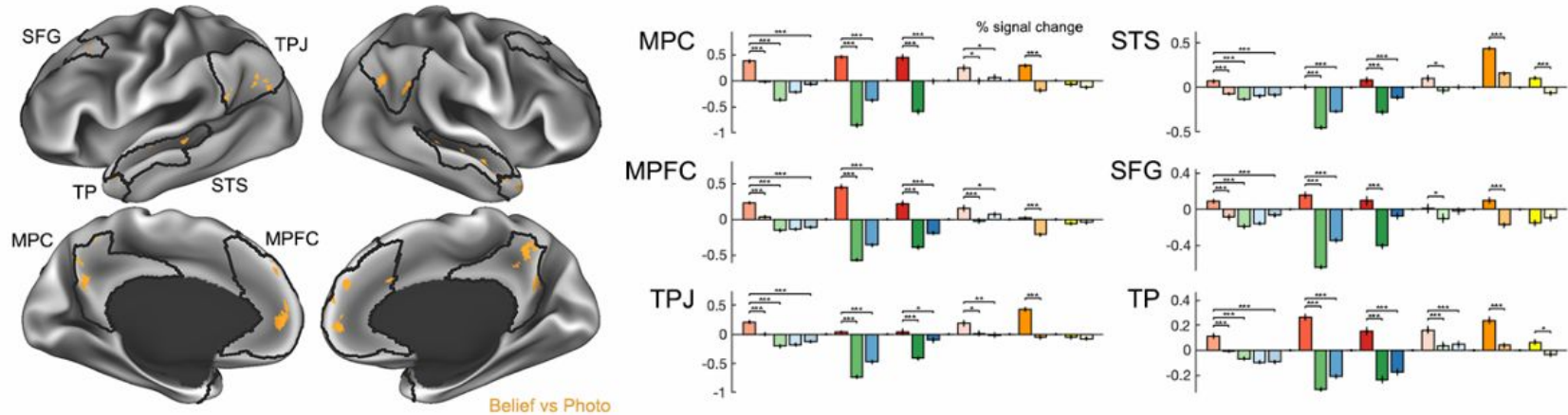


In sum

		MPC	SFG
supplementary motor	postcentral gyrus	precuneus	superior frontal gyrus
			
middle occipital gyrus	superior temporal gyrus SFG	inferior parietal lobe TPJ	the authors'
			

next step

- ROI analysis: using AAL atlas to see if **TP** and **MPFC** will survive correction
- Output figure



Especially thanks to

- Josh's SPM25 tutorials
- NTU's imaging center fMRI workshop
- Those who answered my questions
- Teachers
- TAs
- BH community
- my friends
- lab

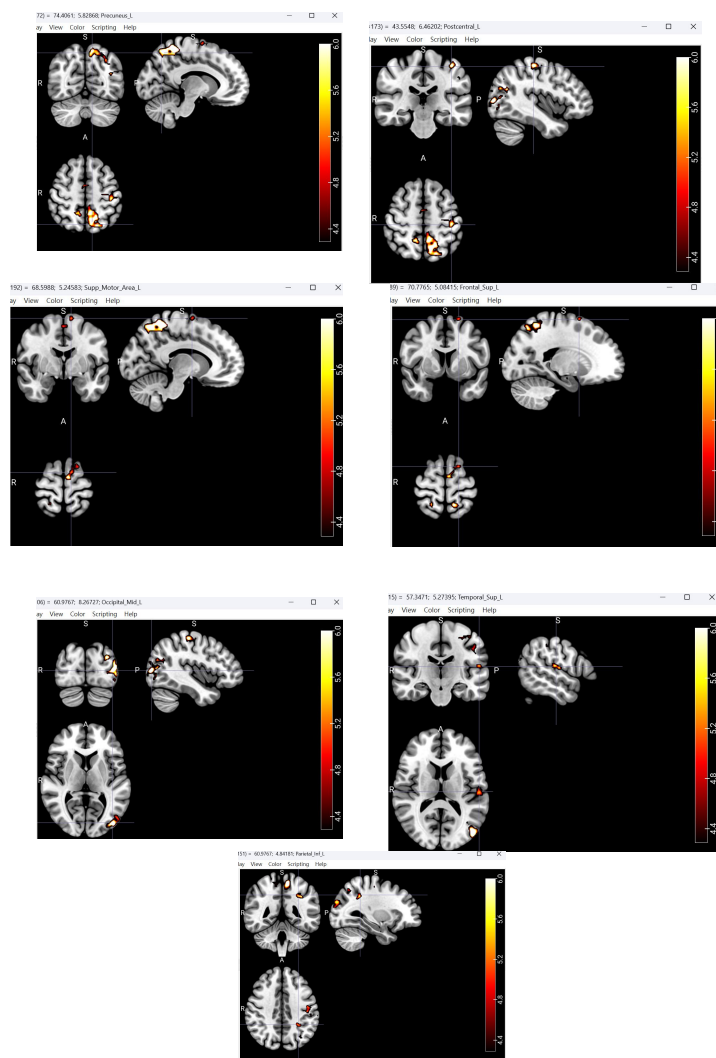
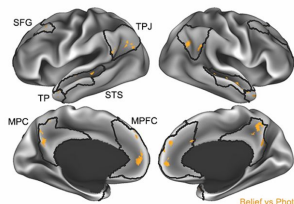
Results

- 7 clusters

- 1. precuneus: perspective taking (MPC)
- 2. postcentral: somatosensory
- 3. supplementary motor area
- 4. superior frontal gyrus: self-awareness (SFG)
- 5. middle occipital gyrus: visual association area
- 6. superior temporal gyrus: social-cognition (STG)
- 7. inferior parietal lobe: social-cognition (TPJ)

- original paper

- medial frontal cortex(MPC)
- medial parietal cortex(MPFC)
- temporo-parietal junction(TPJ)
- superior frontal gyrus(SFG)
- superior temporal sulcus(STS)
- temporal pole(TP)



Useful MRI information

- multiband (3), multiecho (5)
- TR = 2 s
- TE (ms)
 - echo1: 14.4
 - echo2: 33.9
 - echo3: 53.4
 - echo4: 72.9
 - echo5: 92.4
- Voxel resolution: $2.4 \times 2.4 \times 2.5$ mm

Useful MRI information

- number of slices: 48
- slice order:

[0 0.865 1.7325 0.6175 1.485 0.37 1.2375 0.1225 0.99 1.855 0.7425 1.6075
0.495 1.36 0.2475 1.1125 0 0.865 1.7325 0.6175 1.485 0.37 1.2375 0.1225
0.99 1.855 0.7425 1.6075 0.495 1.36 0.2475 1.1125 0 0.865 1.7325 0.6175
1.485 0.37 1.2375 0.1225 0.99 1.855 0.7425 1.6075 0.495 1.36 0.2475
1.1125]

- Experimental design: event-related

MRI acquisition

- TR = 2 s
- TE = 14.4, 33.9, 53.4, 72.9, and 92.4 ms
- $2.4 \times 2.4 \times 2.5$ mm resolution
- 48 oblique axial slices with near whole brain coverage
- multiband acceleration 3x
- GRAPPA acceleration 2x
- interleaved slice acquisition

Preprocessing for 10 subject, too troublesome?

- Create batch for subject 01's preprocessing
- Save batch and script
- Create script for 10 subject's preprocessing

for_loop_version: spm script input

外送代號	你的主控制台傳出的東西	參數檔 _job.m 接收後的任務	格式檢查
inputs{1}	epi_runs_paths{1}	送去 Realign 的 Sess 1 (162 張時間點)	Cell string (162x1)
inputs{2}	epi_runs_paths{2}	送去 Realign 的 Sess 2 (162 張時間點)	Cell string (162x1)
inputs{3}	epi_runs_paths{3}	送去 Realign 的 Sess 3 (162 張時間點)	Cell string (162x1)
inputs{4}	epi_runs_paths{4}	送去 Realign 的 Sess 4 (162 張時間點)	Cell string (162x1)
inputs{5}	{full_t2_path}	送去 Coregister 的 Source & Ref (高畫質 T2)	1x1 Cell 裝著字串
inputs{6}	{full_t1_path}	送去 Coregister 的 Source 與 Segment	1x1 Cell 裝著字串

Step 1~5: Preprocessing

- motion correction
- slice timing
- coregister t2 to epi
- coregister t1 to t2
- segment

- Matlab's SPM in Windows
- Datalad get in Linux

-> expected blocking

-> but not until Dartel create step

Joint Histogram

- Coregistration的derivatives
- 目的: 評估mean EPI 和 T1/T2 之間的對齊程度
- X軸: T1灰階亮度
- Y軸: EPI的灰階亮度
- 圖上每個voxel: 深淺或高度代表多少空間位置同時擁有T1亮度和EPI亮度
- Original joint histogram: 對齊前, mutual information低、entropy (混亂度) 高
- Final joint histogram: 對齊後, SPM透過旋轉、平移將兩張影像結合

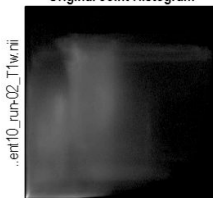
Normalised Mutual Information Coregistration

$$X1 = 0.000^{\circ}X - 1.000^{\circ}Y + 0.002^{\circ}Z + 316.724$$

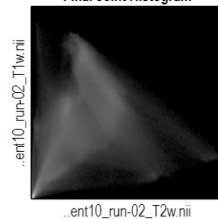
$$Y1 = -0.001^{\circ}X + 0.002^{\circ}Y + 1.000^{\circ}Z + 0.217$$

$$Z1 = 1.000^{\circ}X + 0.000^{\circ}Y + 0.001^{\circ}Z - 0.222$$

Original Joint Histogram



Final Joint Histogram



Check reg: sub01

c1

mean

realign

realign+slice timing

raw

Check reg: sub05

c1

mean

realign

realign+slice timing

raw

Check reg: sub10

c1

mean

realign

realign+slice timing

raw

Original paper

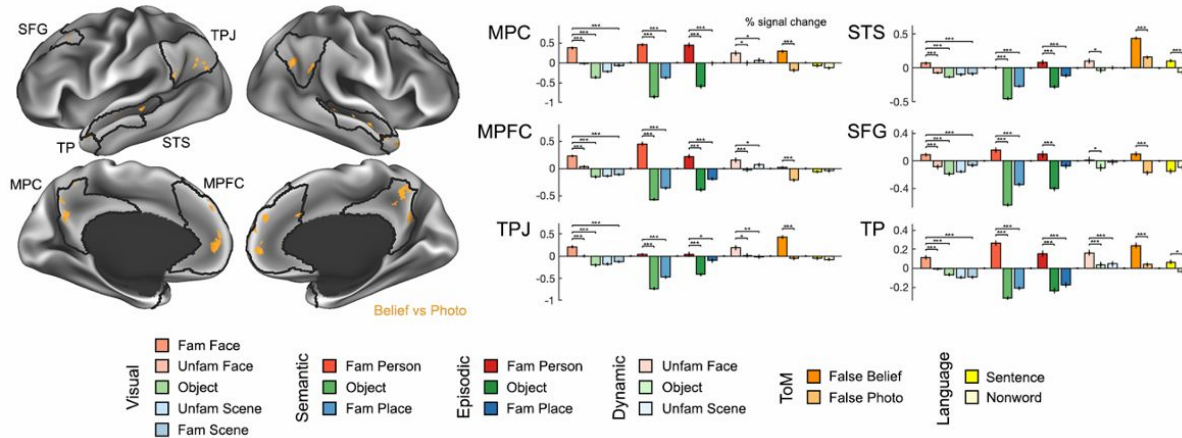
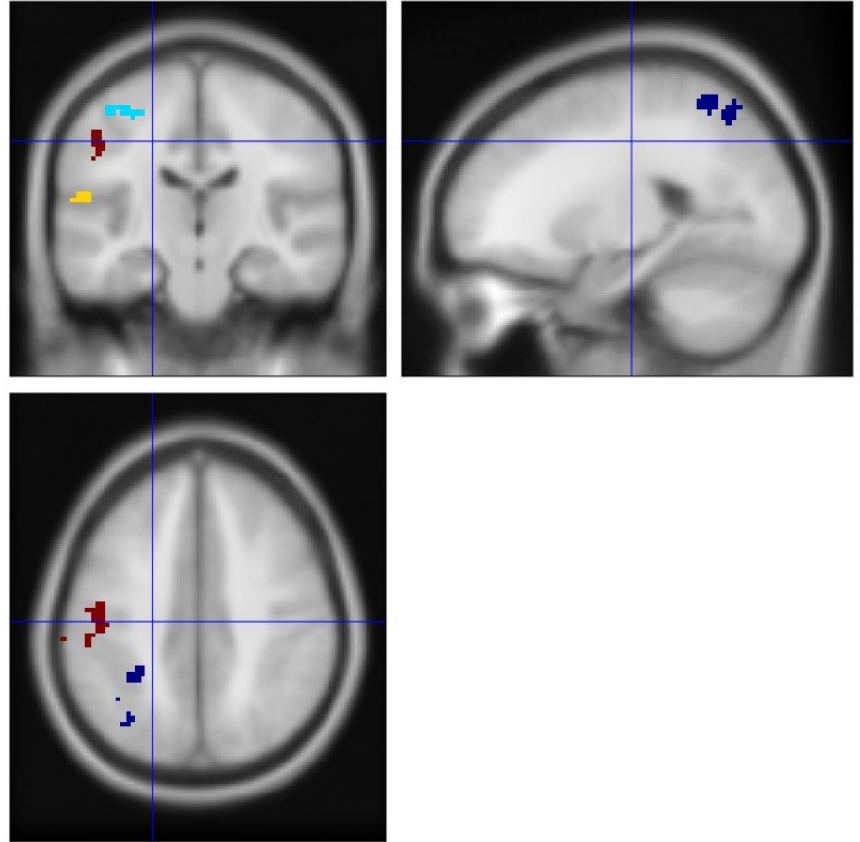


Figure S27: Theory of mind ROI analysis. Left: search spaces and functional ROIs from one representative participant. ROIs were defined as the top 5% of ToM-sensitive coordinates (false belief vs false photo contrast) within each search space. Right: Responses (% signal change) extracted from functionally defined ROIs. Error bars show standard error across runs. * $P < .05/7 = .0071$, ** $P < 10^{-3}$, *** $P < 10^{-4}$ (linear mixed model across runs, with participant included as random effect). Abbreviations: MPC, medial parietal cortex; MPFC, medial prefrontal cortex; SFG, superior frontal gyrus; STS, superior temporal sulcus; TPJ, temporo-parietal junction; TP, temporal pole.

functional-ROI

- MPC_SFG_STG_TPJ
- 20 contrasts
- paired-t test, 10 subjects



functional-ROI

ROI data

Filename: [Not set]

Description: Data from MPC_bin & SFG_bin & STG_bin & TPJ_

Number of time points: 20

Number of regions: 4

Is summarized?: yes

Summary function: mean

Region names:

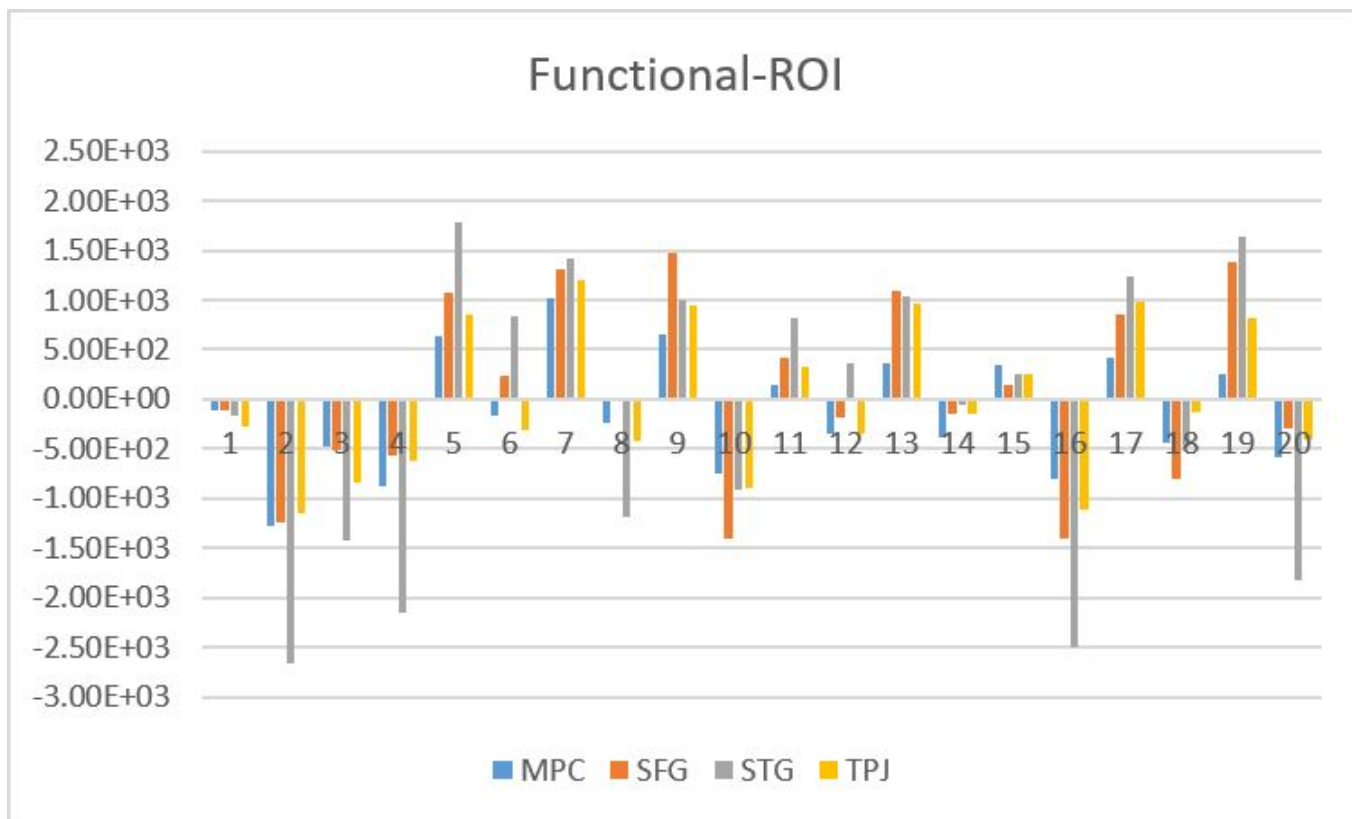
Region 1: MPC_bin

Region 2: SFG_bin

Region 3: STG_bin

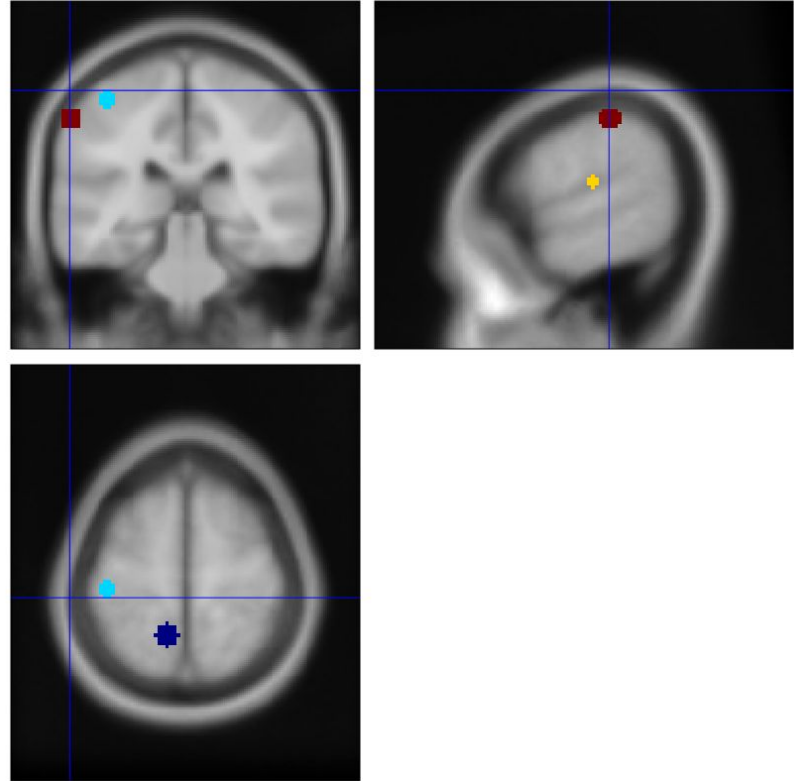
Region 4: TPJ_bin

MPC_SFG_STG_TPJ



Geometric-ROI

- center of sphere:
 - MPC: (-10, -50, 62)
 - SFG: (-41, -26, 58)
 - STG: (-55, -22, 15)
 - TPJ: (-60, -31, 48)
- sphere radius: 6 mm



Geometric-ROI

ROI data

Filename: [Not set]

Description: Data from MPC_sphere & SFG_sphere & STG_sph

Number of time points: 20

Number of regions: 4

Is summarized?: yes

Summary function: mean

Region names:

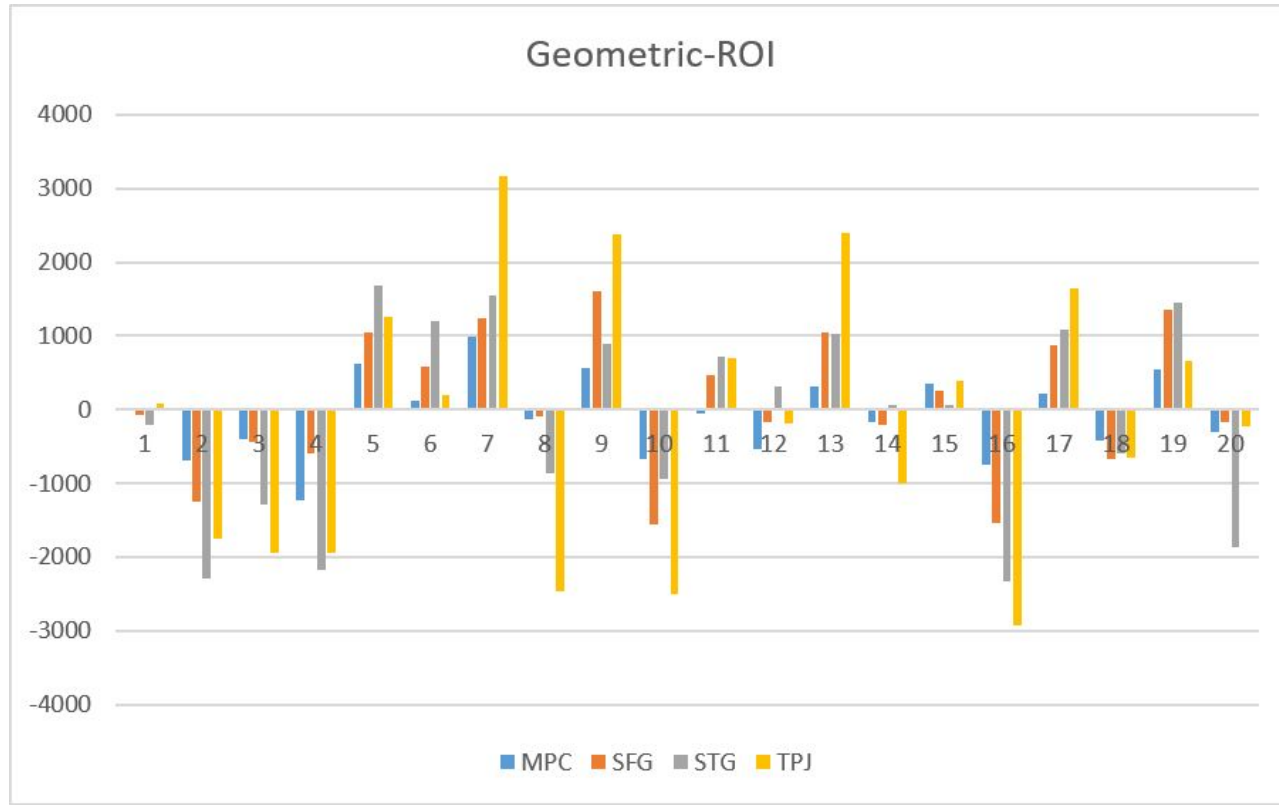
Region 1: MPC_sphere

Region 2: SFG_sphere

Region 3: STG_sphere

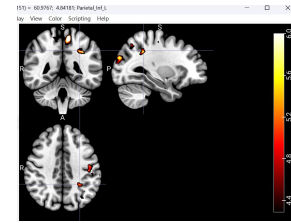
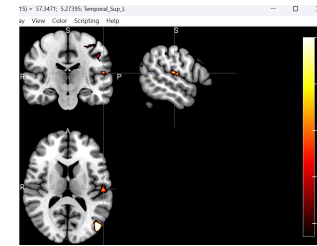
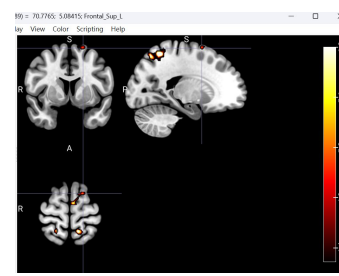
Region 4: TPJ_sphere

Geometric-ROI



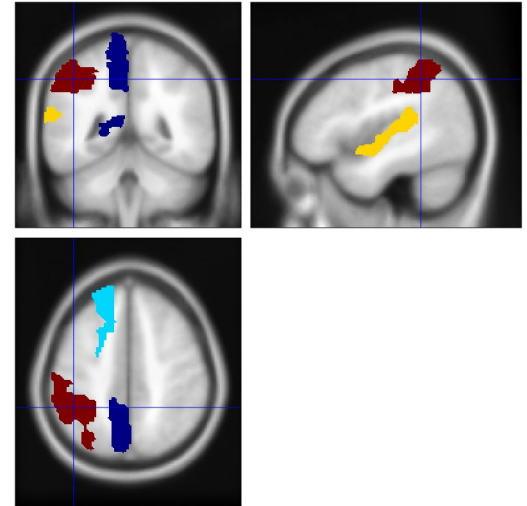
Anatomical-ROI

- 1. precuneus: perspective taking (MPC)
- 2. superior frontal gyrus: self-awareness (SFG)
- 3. superior temporal gyrus: social-cognition (STG)
- 4. inferior parietal lobe: social-cognition (TPJ)



Anatomical-ROI

- 1. precuneus: perspective taking (MPC)
- 2. superior frontal gyrus: self-awareness (SFG)
- 3. superior temporal gyrus: social-cognition (STG)
- 4. inferior parietal lobe: social-cognition (TPJ)

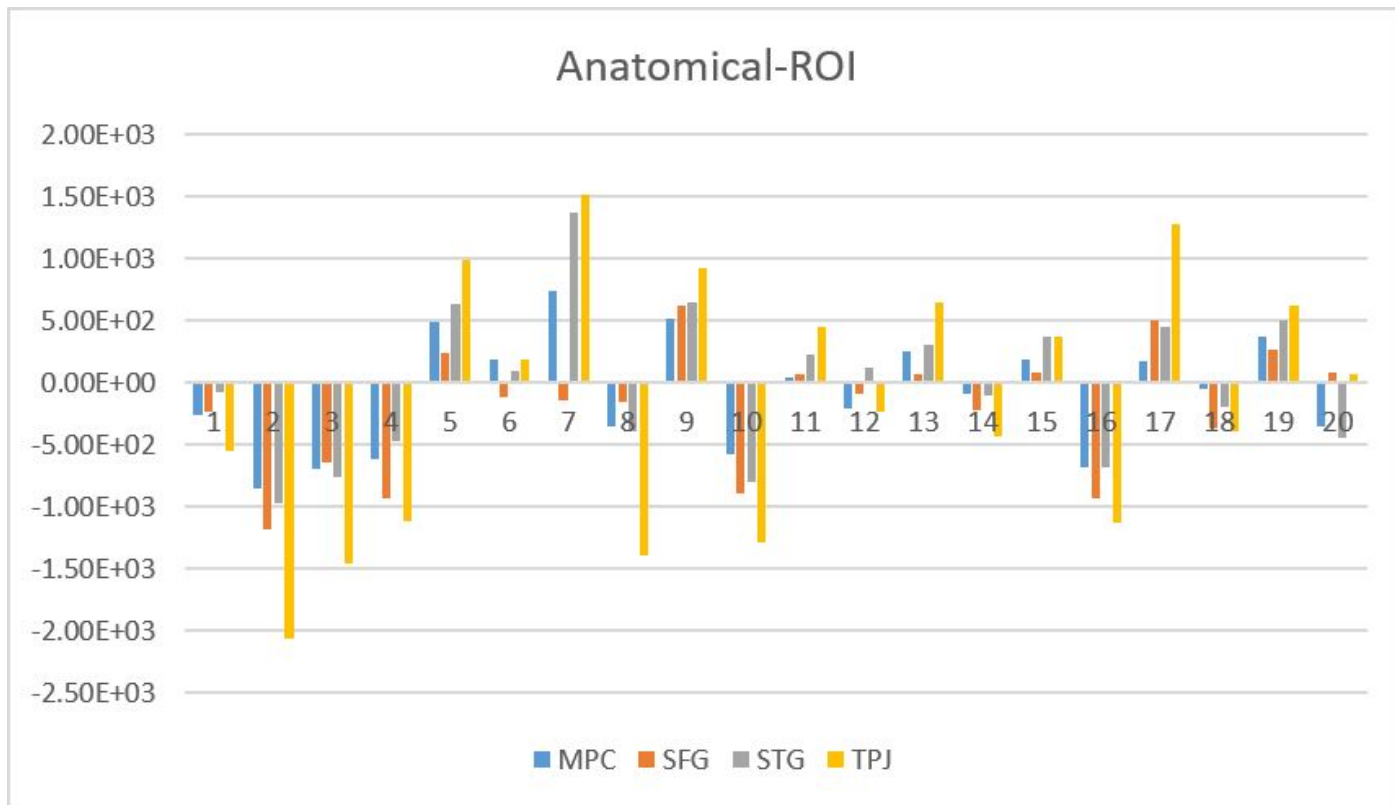


Anatomical-ROI

ROI data

Filename: [Not set]
Description: Data from aal_67 & aal_3 & aal_81 & aal_61
Number of time points: 20
Number of regions: 4
Is summarized?: yes
Summary function: mean
Region names:
Region 1: aal_67
Region 2: aal_3
Region 3: aal_81
Region 4: aal_61

Anatomical-ROI

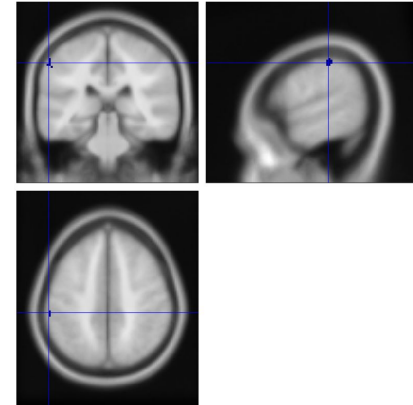
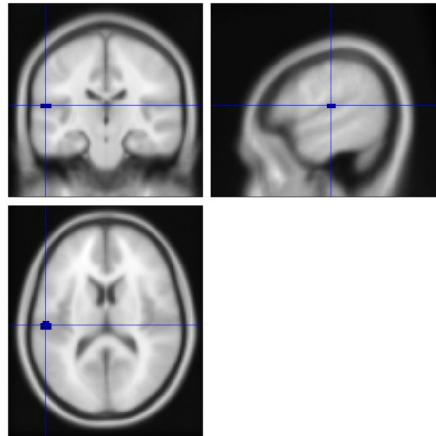
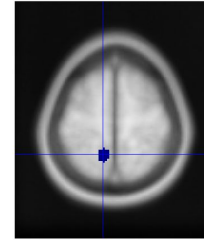
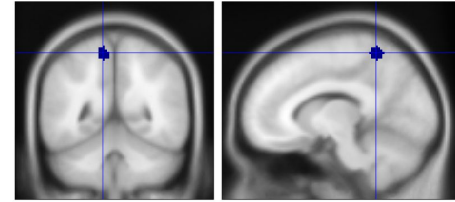


Hybrid-ROI

- intersection of functional+geometric+anatomical ROI
- MPC
- SFG
- STG
- TPJ

Hybrid-ROI

- MPC
- SFG: Warning: No volume resulted for ROI
- STG
- TPJ

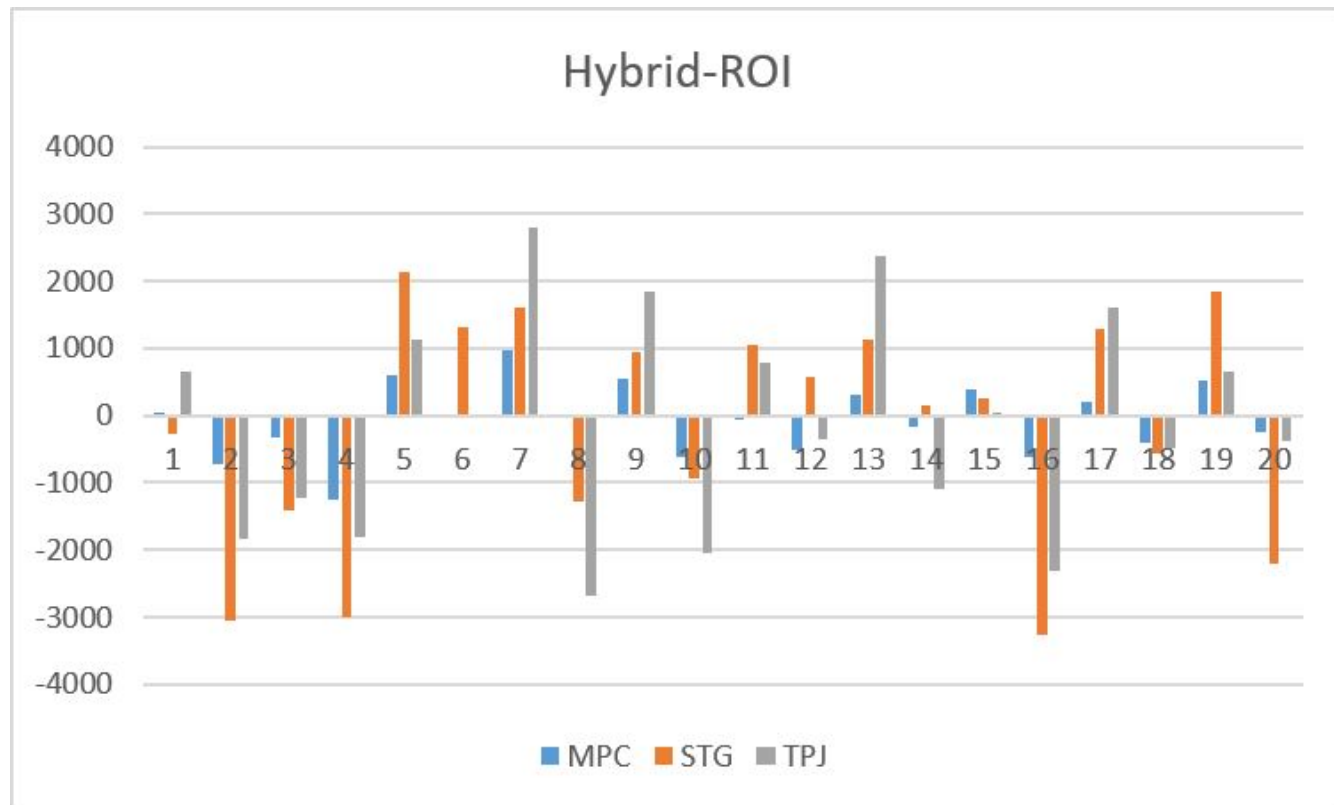


Hybrid-MPC_STG_TPJ

ROI data

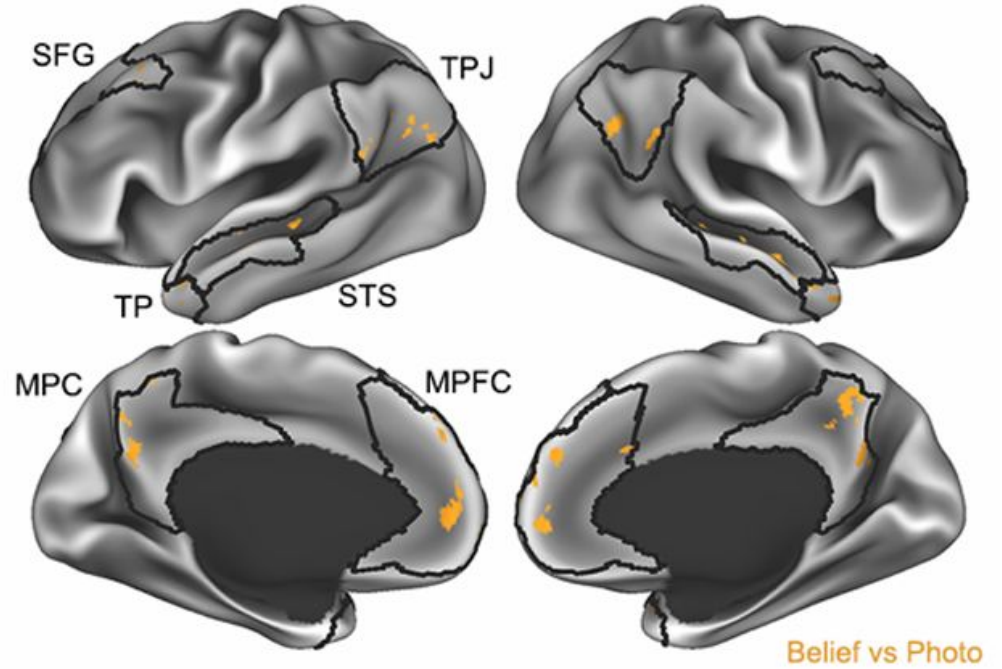
Filename: [Not set]
Description: Data from MPC_FAG & intersection_STG & TPJ_F
Number of time points: 20
Number of regions: 3
Is summarized?: yes
Summary function: mean
Region names:
Region 1: MPC_FAG
Region 2: intersection_STG
Region 3: TPJ_FAG

Hybrid-MPC_STG_TPJ



original paper

- MPFC?
- TPJ?



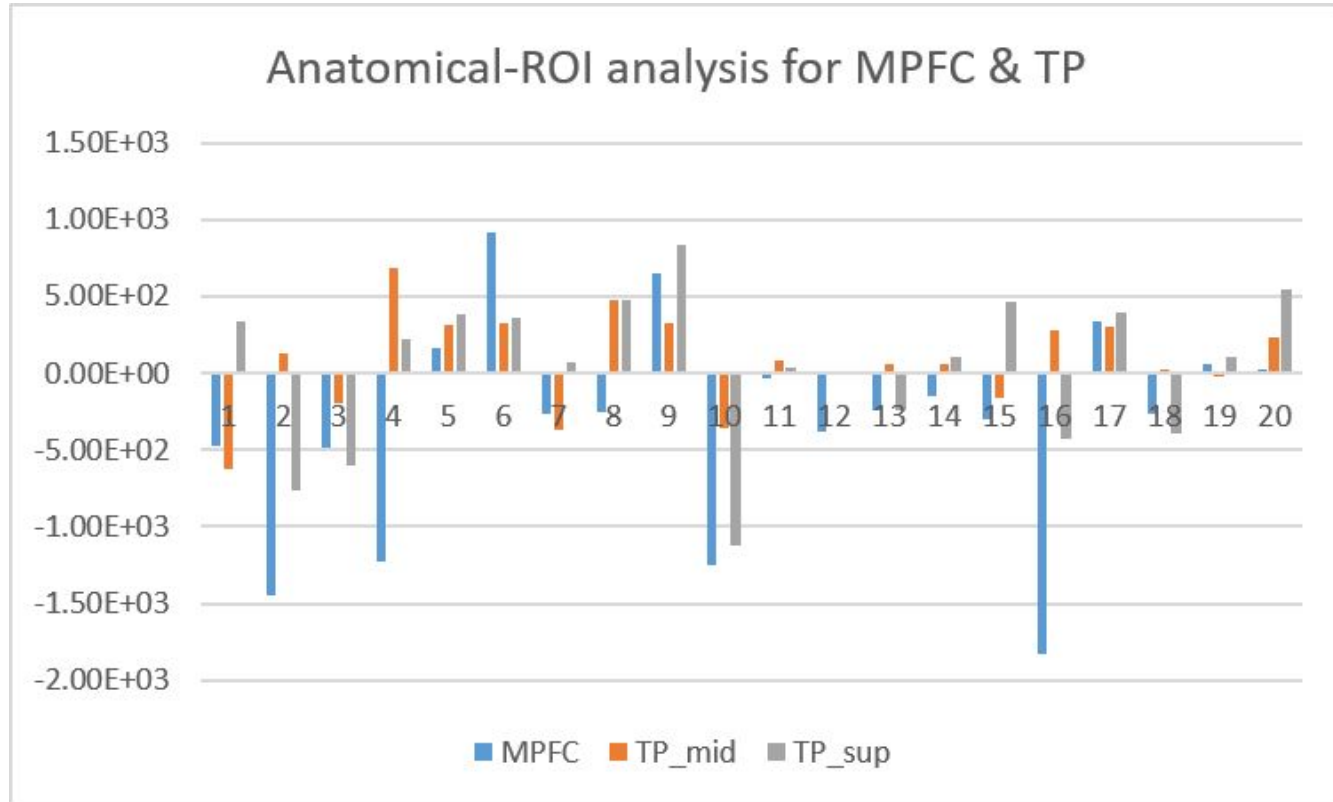
Anatomical-ROI analysis

- all on the left side
- aal23: frontal superior medial L -> medial prefrontal cortex
- aal87: temporal pole middle L -> temporal pole
- aal83: temporal pole superior L -> temporal pole

ROI data

```
Filename: [Not set]
Description:      Data from aal_23 & aal_87 & aal_83
Number of time points: 20
Number of regions: 3
Is summarized?:  yes
Summary function:  mean
Region names:
Region 1:  aal_23
Region 2:  aal_87
Region 3:  aal_83
```

Anatomical-ROI analysis



to find the representative participant...

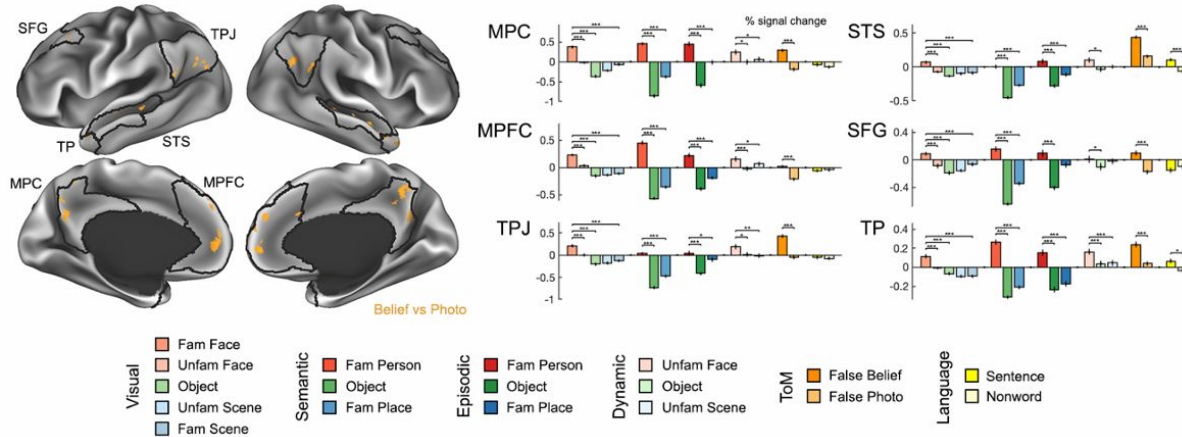


Figure S27: Theory of mind ROI analysis. Left: search spaces and functional ROIs from one representative participant. ROIs were defined as the top 5% of ToM-sensitive coordinates (false belief vs false photo contrast) within each search space. Right: Responses (% signal change) extracted from functionally defined ROIs. Error bars show standard error across runs. * $P < .05/7 = .0071$, ** $P < 10^{-3}$, *** $P < 10^{-4}$ (linear mixed model across runs, with participant included as random effect). Abbreviations: MPC, medial parietal cortex; MPFC, medial prefrontal cortex; SFG, superior frontal gyrus; STS, superior temporal sulcus; TPJ, temporo-parietal junction; TP, temporal pole.

Anatomical-ROI analysis

ROI data

```
Filename: [Not set]
Description:      Data from aal_67 & aal_3 & aal_81 & aal_61 &
Number of time points: 20
Number of regions: 6
Is summarized?: yes
Summary function: mean
Region names:
Region 1:  aal_67
Region 2:  aal_3
Region 3:  aal_81
Region 4:  aal_61
Region 5:  aal_23
Region 6:  aal_83
```

Anatomical-ROI analysis

